



A Biohacker's Guide: Acclimating to CPAP in 90 Days

Evidence-Based Guide to Managing Your Sleep
Apnea With a CPAP Machine

By Dr. Sahil Chopra, MD

Harvard-Trained Sleep Medicine Physician
Co-Founder & Chief Medical Officer, Empower Sleep

MEDICAL DISCLAIMER

This guide is intended for informational and educational purposes only and is not a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified healthcare provider before starting, changing, or discontinuing any supplement, medication, or health regimen.

Some treatments or devices discussed may not be FDA-approved for all sleep disorders.

Individual responses vary, and dosages listed here reflect ranges studied in scientific literature, not personalized medical recommendations.

ABOUT THE AUTHOR



Hello there!

I'm Dr. Sahil Chopra, MD. I'm a quadruple-boarded physician specializing in Internal, Sleep, and Pulmonary/Critical Care Medicine. I completed my Internal Medicine residency at UCLA, Pulmonary/Critical Care fellowship at Loma Linda University, and Sleep Medicine training at Harvard.

We started Empower Sleep with one vision in mind: to provide Harvard-grade sleep care to as many patients as possible. The strategies I adopted in my Sleep Medicine fellowship were deeply rooted in the science and physiology of sleep science. We now apply this science here at Empower Sleep. The goal is to treat all patients as an "n-of-1" (individualized approach). Not a homogeneous, population health-based approach, which is what happens in traditional healthcare. Everyone's sleep is unique to them, as is their sleep disorder(s). When we approach every patient as an individual, we can typically get to the root cause, find a personalized solution, and have stellar clinical outcomes.

At Empower Sleep, we have cared for thousands of patients and analyzed hundreds of thousands of nights of sleep biomarkers. Our innovative care model and success is a by-product of my mentors, my amazing team, our patients, and my family.

Thank you in advance for taking the time to improve your sleep health. I hope you find this guide as a helpful tool to solve your sleep struggles.

It is my wish for all humans to live healthy and fulfilled lives.

Dr. Sahil Chopra, MD



LOMA LINDA UNIVERSITY
HEALTH



01 Why This Guide Exists



The purpose of this guide is to help you make CPAP feel natural and restorative, by walking you through a practical 90-day process designed to build comfort, consistency, and confidence.

When I prescribe CPAP therapy, my goal is never just to hand over a machine in its “factory settings” to fix your Apnea-Hypopnea Index (AHI). My goal is to help you transform your breathing, your sleep, and your overall health. The truth is that CPAP works, but only if the therapy has been **personalized to you**. The first 90 days are the most critical to ensure long-term adherence and therapeutic success.

If you can achieve consistent and comfortable use during these first three months, you’re much more likely to enjoy long-term benefits: reduced cardiovascular risk [1], better blood pressure control [2], improved cognition [3], stabilized mood [4], and hormonal balance [5].

Untreated sleep apnea causes far more than snoring. It is linked to hypertension, heart attacks, stroke, arrhythmias, insulin resistance, and cognitive decline [1–5].

02

CPAP Basics: What You Need to Know Before Starting

How CPAP Works:

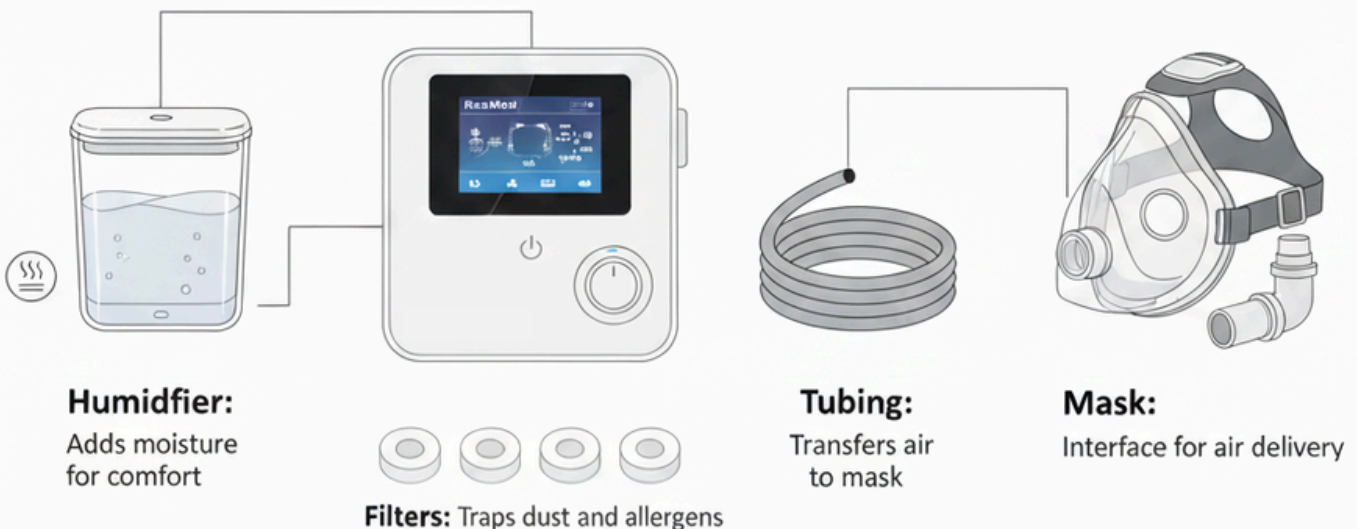
CPAP stands for Continuous Positive Airway Pressure. It delivers a gentle, constant stream of air through a mask to keep your airway open and prevent apneas, a condition where breathing temporarily stops for a period of time [6].

Important: Remember that CPAP machines are not “oxygen” machines. They deliver air under a specified pressure to maintain and open airway during sleep.

Main Components of CPAP Therapy:

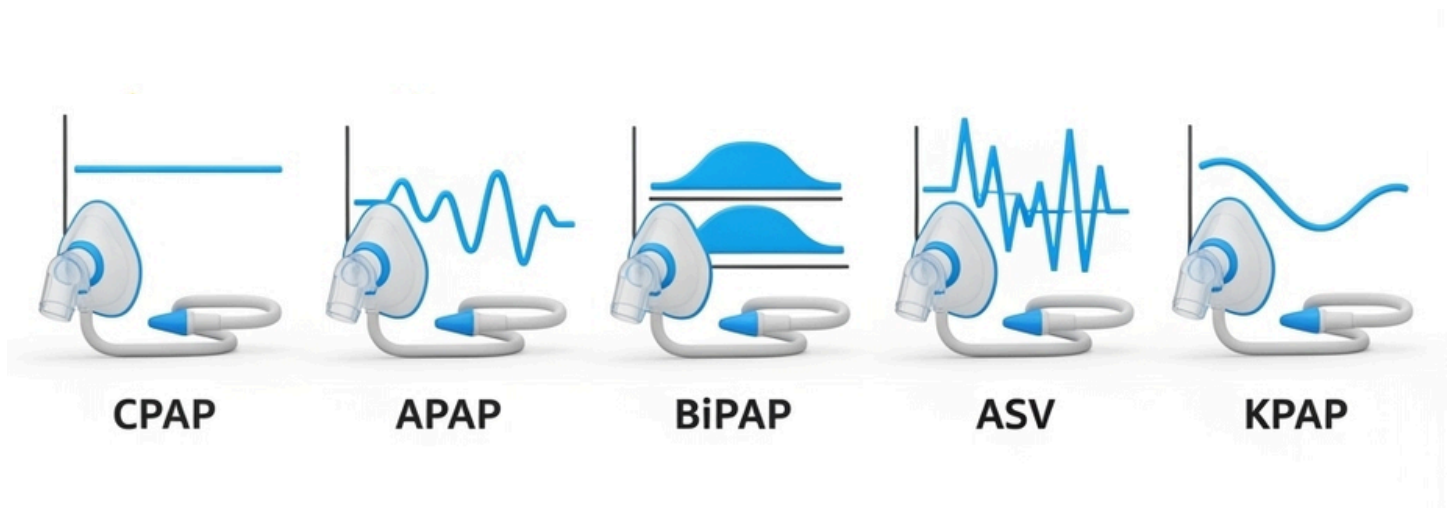
- **CPAP Device** – Think of it like a fan or a blower. It generates the prescribed pressure (or range) to prevent your airway from collapsing
- **Tubing** – Transfer the air from the machine and connect the device to the mask
- **Mask** – The interface that allows the air to be transmitted. There are over 50 different mask configurations (nasal pillows, nasal masks, or full-face options)
- **Humidifier** – Adds moisture to prevent dryness and improve comfort [7]
- **Filters** – Trap dust, allergens, and particles

CPAP Device:



Types of Positive Airway Pressure (PAP) Therapies:

- **Continuous Positive Airway Pressure (CPAP):** A stream of pressure that is *fixed* and continuous
- **Automatic Positive Airway Pressure (APAP):** This is the **most commonly prescribed form of PAP**. It can auto-adjust based on detected airway resistance. **The terms APAP and CPAP are often used interchangeably**. All modern day APAP machines are capable of being *fixed* CPAP machines
- **Bilevel Positive Airway Pressure (BiPAP):** Provides higher pressure during inhalation and lower during exhalation, often for more complex breathing disorders [8]. BiPAP machines come in a fixed and auto-adjusting configuration
- **Adaptive Servo-Ventilation (ASV):** Continuously monitors breathing patterns and dynamically adjusts both inspiratory and expiratory pressures on a breath-by-breath basis. Primarily used for central sleep apnea, Cheyne–Stokes respiration, or complex sleep apnea [9]
- **Kairos (KPAP):** This was invented recently by a sleep doctor (Dr. William Noah) . These devices are coming soon and are in the FDA clearance process. Personally, I'm excited about these machines. KPAP dynamically modulates inspiratory and expiratory pressures to improve comfort. The goal is for these machines to be more comfortable and synchronous with our breathing, thereby improving patient outcomes



03 The 90-Day Roadmap—6 phases

Note: For pragmatic reasons you can replace the term CPAP with the type of machine you use. This guide is applicable in most scenarios.

Each phase has specific goals, action steps, and troubleshooting tips.

Phase 1: Days 1–15 *Familiarization & Setup*

Goals:

- Wear your mask daily, even if only for short periods
- Begin nightly CPAP use, even if not for the entire night

Actions:

- Practice wearing mask during calm, non-sleep activities (reading, watching TV)
- Use “ramp” mode to start at lower pressure [9]
- Adjust humidification to prevent dryness [7]

Troubleshooting:

- For claustrophobia, start with short “practice breathing” sessions while awake
- For nasal dryness, add heated humidification or use saline spray before bed [7]

Phase 2: Days 16–30 *Build Tolerance*

Goals:

- Reach ≥ 4 hours per night of CPAP use on most nights
- Begin tracking data from your machine (AHI and leak rate) and your sleep test
- I would strongly focus on how you’re subjectively feeling and your objective breathing (AHI estimated by the Empower Sleep ring or lab study)

Actions:

- Check your Empower Sleep app, PAP device app, or download your usage data (SD card) weekly [11]
- Ensure straps are snug but not overly tight
- Continue daytime practice if discomfort persists
- Try expiratory pressure relief (EPR) features to make exhalation more natural [16]

Troubleshooting:

- If air swallowing (aerophagia) occurs, slow the ramp increase or lower the pressure settings with your physician/coach [12]

03 The 90-Day Roadmap—6 phases

Phase 3: Days 31–45 *Establishing Consistency*

Goals:

- Use CPAP for the entire sleep period most nights
- Resolve persistent comfort issues

Actions:

- Assess whether your mask style is optimal; switch if needed
- Increase humidifier settings if you live in a dry climate [7]
- Use heated tubing to prevent condensation (“rainout”) [13]

Troubleshooting:

- For skin irritation, use mask liners or memory foam cushions [14]
- For nasal congestion, ensure humidity is sufficient and keep filters clean [7,15]. You can also consider using OTC saline or steroid nasal sprays as a mechanism for nasal decongestion

Phase 4: Days 46–60 *Optimizing Comfort*

Goals:

- Make CPAP feel like a normal part of your routine
- Address fine-tuning issues

Actions:

- Maintain optimal bedroom environment (cool, dark, and quiet)
- Maintain a consistent sleep schedule

Troubleshooting:

- If you wake with dry mouth and you are noticing the water chamber is empty, it is a clue that there could be a substantial leak. Consider using a different mask, chin strap or mouth tape to help [17]. Check in with your provider and/or sleep coach about this if the above does not resolve your symptoms

03 The 90-Day Roadmap–6 phases

Phase 5: Days 61–75 *Troubleshooting & Long-Term Prep*

Goals:

- Resolve any ongoing barriers
- Prepare for travel or unusual sleep environments

Actions:

- Consider travel CPAP machines and FAA-approved battery backups [18]
- Reorder supplies: cushions, filters, and tubing [19]
- Keep up with regular cleaning schedule [19]

Troubleshooting:

- If the machine noise disturbs your partner, place the device on a soft mat or in a position lower than your bed [20]

Phase 6: Days 76–90 *Transition to Lifelong Use*

Goals:

- Consistent full-night use without discomfort
- Maintenance plan in place

Actions:

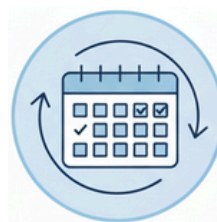
- Review your 90-day usage data with your physician and/or sleep coach
- Schedule filter and mask replacements in advance
- Plan for annual or 6 monthly follow-up sleep evaluations



PHASE 1



PHASE 2



PHASE 3



PHASE 4



PHASE 5



PHASE 6

04

COMMON BARRIERS & SOLUTIONS—EXPANDED

Technical Issues:

- **Mask leaks:** Readjust fit, try alternate mask style, and replace worn cushions [21]
- **Pressure Intolerance:** Use ramp, enable pressure relief, and review pressure ranges [12]
- **Rainout:** Use adjust humidity and try heated tubing and/or insulate hose [13]

Physical Comfort:

- **Dryness:** Increase humidity, use saline spray, and address mouth leaks [7]
- **Skin irritation:** Clean mask daily, use liners, and rotate mask types [14]

Psychological & Motivation:

- **Claustrophobia:** Gradually increase exposure time while awake [10]
- **Stigma:** Reframe CPAP as a longevity and performance-enhancing tool [1–5]

Symptom	Likely Cause	Solution	Ref
 Dry mouth	Mouth leak	Chin strap or full-face mask	17
 Nasal congestion	Low humidity	Increase humidifier, heated tubing	7, 13
 Mask marks	Overtight straps	Loosen straps, add liner	14

05

CPAP MAINTENANCE & SUPPLY MANAGEMENT

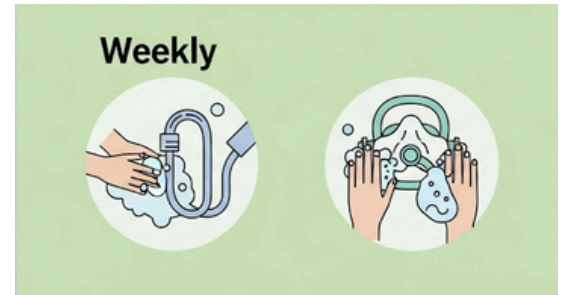
Daily:

- Empty humidifier chamber and rinse with distilled water
- Wipe mask cushion



Weekly:

- Wash tubing, mask, and humidifier chamber with mild soap [19]

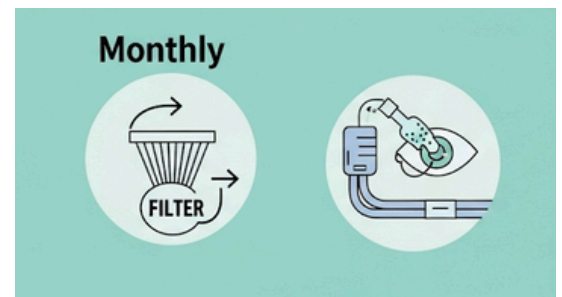


Monthly:

- Replace disposable filters
- Inspect tubing for cracks [19]

Replacement Guidelines (Medicare/insurance typical):

- **Mask cushions:** every month
- **Tubing:** every 3 months
- **Headgear:** every 6 months
- **Filters:** monthly [19]



06

MEASURING PROGRESS & STAYING MOTIVATED

Objective markers:

- From CPAP Machine: AHI <5/hr, leak <24 L/min, ≥ 4 hours/night [22]
- From Empower Sleep Testing: Sleep quality index to be higher than original baseline. The goal is different for everyone. sAHI < 10/hr.

Subjective markers:

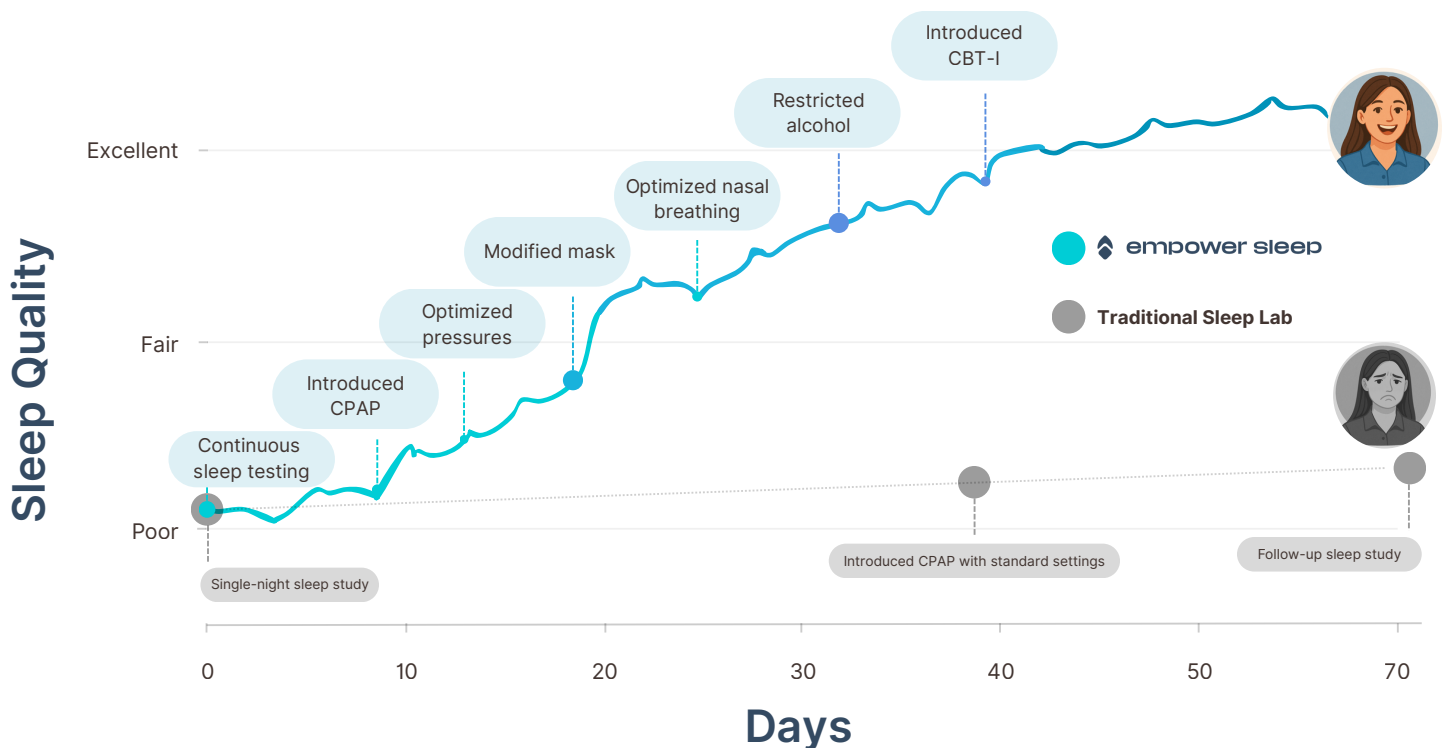
- Less daytime sleepiness, improved mood, better blood pressure control, and more energy [1–4].

Tracking your progress via manufacturer apps or telemonitoring not only motivates you but also allows for earlier troubleshooting [11].

empower sleep vs Traditional Sleep Labs

Longitudinal Sleep Testing (Continuous)

Single-Night Sleep Testing (Snapshot)



07

CPAP & YOUR HEALTH—THE BIGGER PICTURE

Adherence to CPAP has been linked to:

- Reduced cardiovascular and stroke risk [1]
- Lowered blood pressure [2]
- Improved cognitive function and slower decline [3]
- Better insulin sensitivity and metabolic health [5]

08

TRAVEL & LIFESTYLE INTEGRATION

- Always carry your CPAP as a medical device in your hand luggage. It's a medical device and exempt from baggage limits [18]
- Use travel-sized devices and FAA-approved batteries
- Keep a minimal cleaning kit for trips



09

CONCLUSION – YOUR NEXT STEPS

Your first 90 days with CPAP are your launchpad. Follow this phased roadmap, address barriers quickly, and let CPAP become a seamless part of your nights. The health benefits—for your heart, brain, and quality of life—are well worth the effort.

Next Step: Book your 30-day and 90-day check-ins at www.empowersleep.com to keep your progress on track.

Get Personalized Sleep Care, Guided By Experts

Our virtual sleep care program uses your data to create a plan that works for you

www.empowersleep.com

IF YOU HAVE ANY QUESTIONS, MESSAGE US ON OUR SOCIAL CHANNELS.
WE CAN MAKE A VIDEO TO ANSWER YOUR QUESTIONS.

   @empowersleep

You could also email us at care@empowersleep.com

1. Peker Y, Glantz H, Eulenburg C, Wegscheider K, Herlitz J, Thunström E. Effect of positive airway pressure on cardiovascular outcomes in coronary artery disease patients with nonsleepy obstructive sleep apnea: The RICCADSA randomized controlled trial. *Am J Respir Crit Care Med*. 2016;194(5):613-20. doi:10.1164/rccm.201601-0088OC. PMID: 26914592.
2. Haentjens P, Van Meerhaeghe A, Moscariello A, Poppe K, Dupont A, Gillebert TC, et al. The impact of continuous positive airway pressure on blood pressure in patients with obstructive sleep apnea syndrome: Evidence from a meta-analysis of placebo-controlled randomized trials. *Arch Intern Med*. 2007;167(8):757-64. doi:10.1001/archinte.167.8.757. PMID: 17452537.
3. Richards KC, Gooneratne NS, Dicicco B, Hanlon A, Moelter S, Onen F, et al. CPAP adherence may slow 1-year cognitive decline in older adults with mild cognitive impairment and apnea. *J Am Geriatr Soc*. 2019;67(3):558-64. doi:10.1111/jgs.15616. PMID: 30697872.
4. Schwartz DJ, Kohler WC, Karatinos G. Symptoms of depression in individuals with obstructive sleep apnea may be amenable to treatment with continuous positive airway pressure. *Chest*. 2005;128(3):1304-9. doi:10.1378/chest.128.3.1304. PMID: 16162713.
5. Grunstein RR, Handelsman DJ, Lawrence SJ, Blackwell C, Caterson ID, Sullivan CE. Neuroendocrine dysfunction in sleep apnea: reversal by continuous positive airways pressure therapy. *J Clin Endocrinol Metab*. 1989 Feb;68(2):352-8. doi: 10.1210/jcem-68-2-352. PMID: 2493027.
6. Pinto VL, Sankari A, Sharma S. Continuous Positive Airway Pressure. [Updated 2025 Jul 7]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-.
7. Massie CA, Hart RW, Peralez K, Richards GN. Effects of humidification on nasal symptoms and compliance in sleep apnea patients using continuous positive airway pressure. *Chest*. 1999;116(2):403-8. doi:10.1378/chest.116.2.403. PMID: 10453869.
8. Kushida CA, Chediak A, Berry RB, Brown LK, Gozal D, Iber C, et al. Clinical guidelines for the manual titration of positive airway pressure in patients with obstructive sleep apnea. *J Clin Sleep Med*. 2008;4(2):157-71. PMID: 18468315.
9. Tomita Y, Kasai T. Effectiveness of adaptive servo-ventilation. *World Journal of Respiriology*. 2015;5(2):112-125. doi:10.5320/wjr.v5.i2.112.
10. Means MK, Edinger JD, Husain AM. CPAP compliance in sleep apnea patients with and without claustrophobic tendencies. *Sleep Med*. 2004;5(5):425-31. doi:10.1016/j.sleep.2004.01.017. PMID: 15336233.
11. Stepnowsky CJ, Palau JJ, Gifford AL, Ancoli-Israel S. A self-management approach to improving continuous positive airway pressure adherence and outcomes. *Behav Sleep Med*. 2007;5(2):131-46. doi: 10.1080/15402000701190622. PMID: 17441783.

12. Shepherd K, Hillman D, Eastwood P. Symptoms of aerophagia are common in patients on continuous positive airway pressure therapy and are related to the presence of nighttime gastroesophageal reflux. *J Clin Sleep Med*. 2013 Jan 15;9(1):13-7. doi: 10.5664/jcsm.2328. PMID: 23319899; PMCID: PMC3525982.
13. Nilius G, Domanski U, Franke KJ, Rühle KH. Impact of a controlled heated breathing tube humidifier on sleep quality during CPAP therapy in a cool sleeping environment. *Eur Respir J*. 2008 Apr;31(4):830-6. doi: 10.1183/09031936.00161806. Epub 2008 Jan 9. PMID: 18184678.
14. Bachour A, Avellan-Hietanen H, Palotie T, Virkkula P. Practical Aspects of Interface Application in CPAP Treatment. *Can Respir J*. 2019 Nov 3;2019:7215258. doi: 10.1155/2019/7215258. PMID: 31781315; PMCID: PMC6874967.
15. Ortolano GA, Schaffer J, McAlister MB, Stanchfield I, Hill E, Vandenburg L, Lewis M, John S, Canonica FP, Cervia JS. Filters reduce the risk of bacterial transmission from contaminated heated humidifiers used with CPAP for obstructive sleep apnea. *J Clin Sleep Med*. 2007 Dec 15;3(7):700-5. PMID: 18198803; PMCID: PMC2556912.
16. Sleeper GP, Catullo KA, Chen PL, Patil SP, Strohl KP, Chiang AA. The influence of an innovative inspiratory pressure relief accessory on continuous positive airway pressure profiles in a laboratory bench setting. *J Clin Sleep Med*. 2025 Jun 1;21(6):1073-1079. doi: 10.5664/jcsm.11624. PMID: 40078098; PMCID: PMC12134585.
17. Matsumura E, Grad GF, Madeiro F, Genta PR, Lorenzi-Filho G. Mouth Leak Is Associated with Sleep Fragmentation During Nasal Continuous Positive Airway Pressure Treatment of Obstructive Sleep Apnea and May Be Detected by Leak Waveform Analysis. *Ann Am Thorac Soc*. 2025 Mar;22(3):430-437. doi: 10.1513/AnnalsATS.202404-425OC. PMID: 39499768.
18. Bodington R, Johnson O, Carveth-Johnson P, Faruqi S. Travel with CPAP machines: how frequent and what are the problems? *J Travel Med*. 2018 Jan 1;25(1). doi: 10.1093/jtm/tax085. PMID: 29232460.
19. ResMed. How to clean your CPAP equipment. ResMed. n.d. <https://www.resmed.com/en-us/sleep-apnea/cpap-parts-support/cleaning-cpap-equipment/>
20. UCLA Health. Living with Positive Airway Pressure (PAP). 2025. Available from: <https://www.uclahealth.org/medical-services/sleep-medicine/patient-resources/patient-education/living-with-pap>
21. Rühle KH, Randerath W. Erfassung des Maskenlecks unter CPAP bei Patienten mit obstruktivem Schlafapnoesyndrom [Measurement of mask leakage during CPAP in patients with obstructive sleep apnea]. *Pneumologie*. 2000 Oct;54(10):422-4. German. doi: 10.1055/s-2000-7688. PMID: 11089392.
22. Weaver TE, Grunstein RR. Adherence to continuous positive airway pressure therapy: the challenge to effective treatment. *Proc Am Thorac Soc*. 2008 Feb 15;5(2):173-8. doi: 10.1513/pats.200708-119MG. PMID: 18250209; PMCID: PMC2645251.